



✧ Activity 3

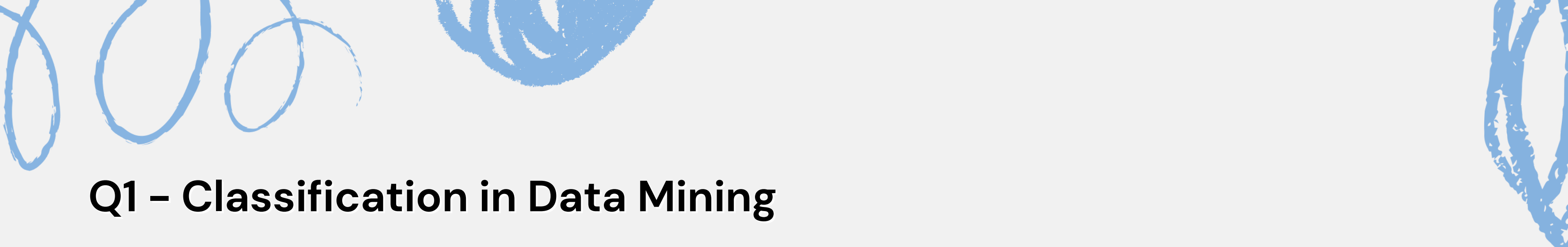
Data Mining

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Q1 – Classification in Data Mining

Classification is a data mining technique used to predict group membership for data instances. It is a form of supervised learning where a model is trained on a labeled dataset (i.e., data with known class labels), and then used to classify new, unseen instances into predefined categories or classes.

For example, a classification algorithm could be used to predict whether an email is spam or not based on its content.

Source:

Han, J., Pei, J., & Kamber, M. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann.



Q2 – The Importance of Classification in Data Mining

Predictive Modeling	Classification is used to build models that can predict the class labels of future or unknown data instances.
Enhanced Decision Making	It supports decision-making processes by providing accurate forecasts based on historical data.
Improved Data Understanding	Helps users understand the structure and distribution of data through clearly defined classes.
Automation of Business Processes	Automates tasks such as spam filtering, loan approval, and medical diagnosis, leading to time and cost efficiency.
Supports Strategic Planning	Enables organizations to plan marketing campaigns, risk assessments, and customer retention strategies based on predictive insights.

Source:

Han, J., Pei, J., & Kamber, M. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann.

Real-World Applications of Classification in Data Mining

- Healthcare – Doctors can use data classification for numerous reasons. For example, they can group certain indicators of a disease for improved diagnostics. Likewise, classification comes in handy when grouping patients by age, condition, and other key factors.
- Finance – Data classification is essential for financial institutions. Banks can group information about consumers to find lenders more easily. Furthermore, data classification is crucial for elevating security.
- E-commerce – A key feature of online shopping platforms is recommending your next buy. They do so with the help of data classification. A system can analyze your previous decisions and group the related information to enhance recommendations.
- Weather forecast – Several considerations come into play during a weather forecast, including temperatures and humidity. Specialists can use a data mining platform to classify these considerations.

Source:

<https://www.opit.com/magazine/classification-in-data-mining/>



Q3 – Classification Algorithms

- **Logistic Regression**
- Naive Bayes
- Artificial neural network
- **Decision tree**
- Binary classification
- KNN
- Ensemble learning
- **Clustering**
- Random forest
- Multiclass classification
- Support Vector Machine

Source:

<https://www.google.com/search>



Q3 – Logistic Regression Classification Algorithms

Logistic regression is a supervised machine learning algorithm that accomplishes binary classification tasks by predicting the probability of an outcome, event, or observation. The model delivers a binary or dichotomous outcome limited to two possible outcomes: yes/no, 0/1, or true/false.

Binary logistic regression	Multinomial logistic regression	Ordinal logistic regression
Used when the outcome has only two categories (e.g., Yes/No, Pass/Fail). It's a simple either/or case.	Used when there are more than two categories, but no natural order between them. Think of multiple options with equal status.	Used when there are more than two categories that are ranked or ordered, like sizes or levels.
1.Deciding on whether or not to offer a loan to a bank customer: Outcome = yes or no. 2.Evaluating the risk of cancer: Outcome = high or low. 3.Predicting a team's win in a football match: Outcome = yes or no.	1.Predicting whether a student will join a college, vocational/trade school, or corporate industry. 2.Estimating the type of food consumed by pets, the outcome may be wet food, dry food, or junk food.	1.Formal shirt size: Outcomes = XS/S/M/L/XL 2.Survey answers: Outcomes = Agree/Disagree/Unsure 3.Scores on a math test: Outcomes = Poor/Average/Good

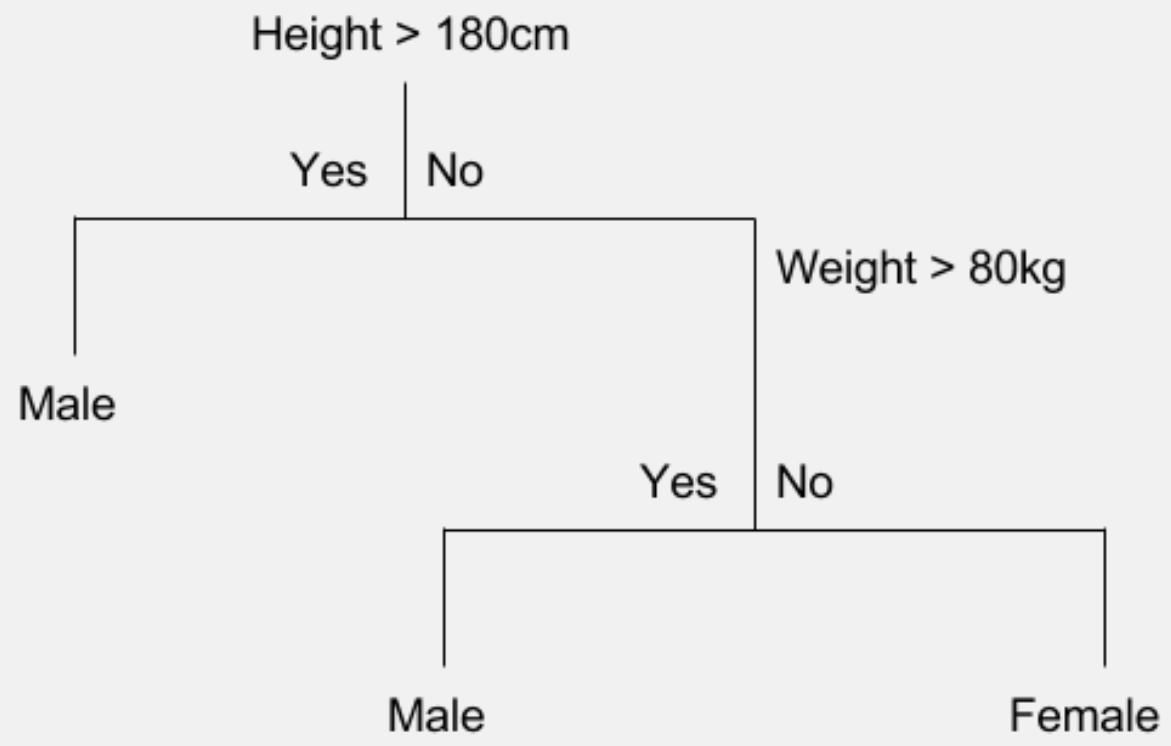
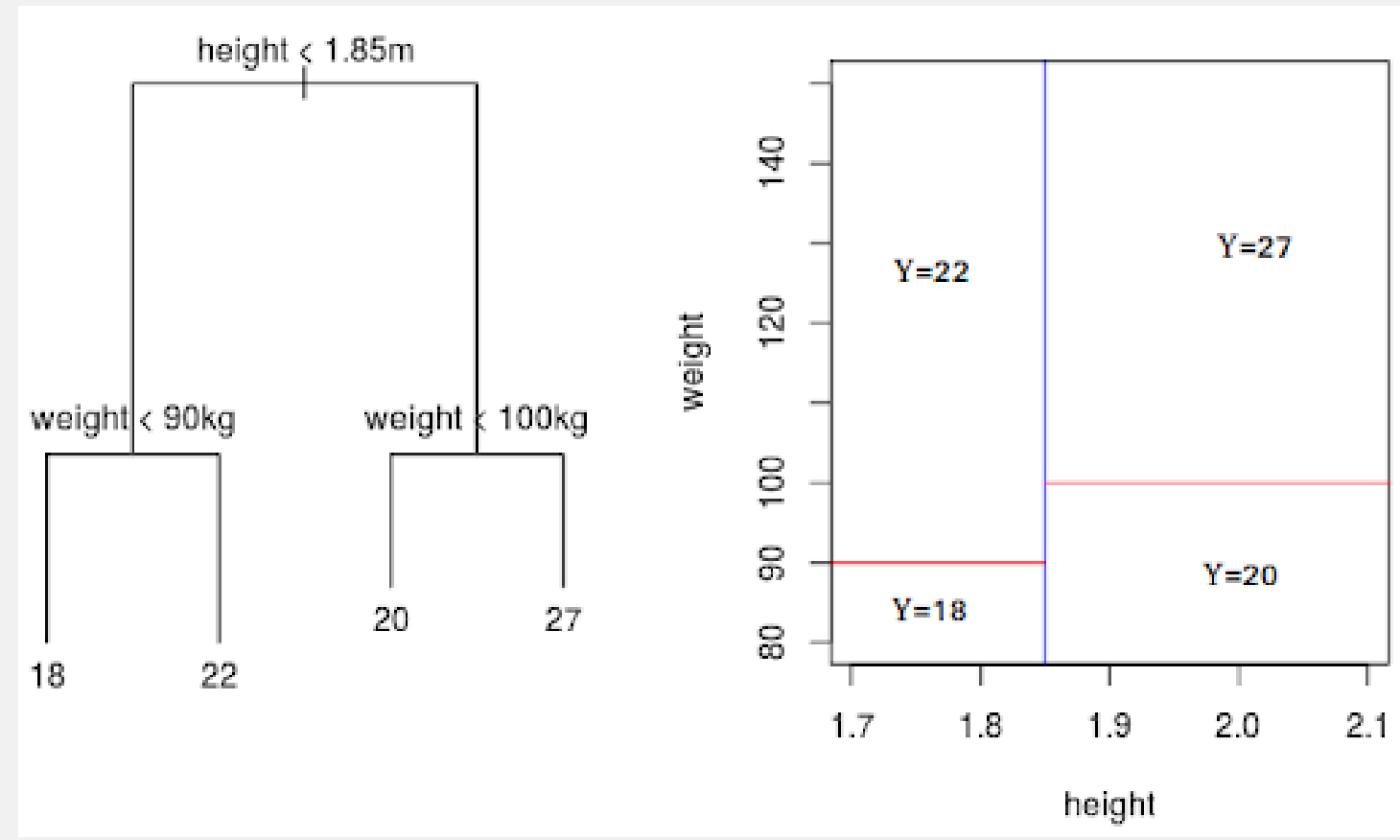
Source:

<https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-logistic-regression/>

Q3 – Decision tree Classification Algorithms

A decision tree is a supervised machine learning algorithm used for classification and regression tasks. Decision trees are tree data structures that continuously split data from a given set of observations according to a certain parameter.

- classification trees (categorical/qualitative outcomes)
- regression trees (continuous/quantitative outcomes).

Classification Trees (Yes/No Types)	Regression Trees (Continuous Data Types)
	

Source:

<https://builtin.com/data-science/classification-tree>

Q3 – Clustering Classification Algorithms

Clustering is an unsupervised machine learning technique designed to group unlabeled examples based on their similarity to each other.

Clustering is useful in a variety of industries. Some common applications for clustering:

- Market segmentation
- Social network analysis
- Search result grouping
- Medical imaging
- Image segmentation
- Anomaly detection

Some specific examples of clustering:

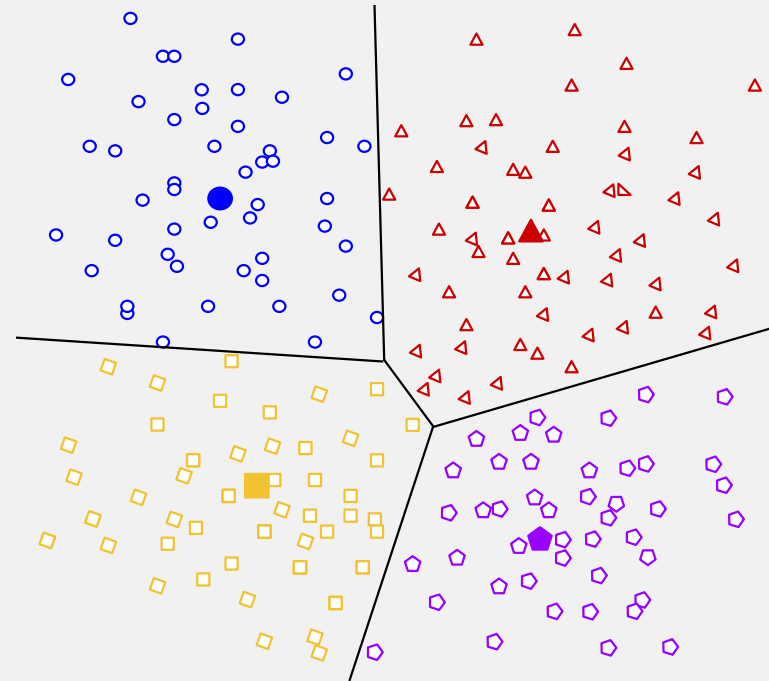
- The Hertzsprung–Russell diagram shows clusters of stars when plotted by luminosity and temperature.
- Gene sequencing that shows previously unknown genetic similarities and dissimilarities between species has led to the revision of taxonomies previously based on appearances.
- The Big 5 model of personality traits was developed by clustering words that describe personality into 5 groups. The HEXACO model uses 6 clusters instead of 5.

Source:

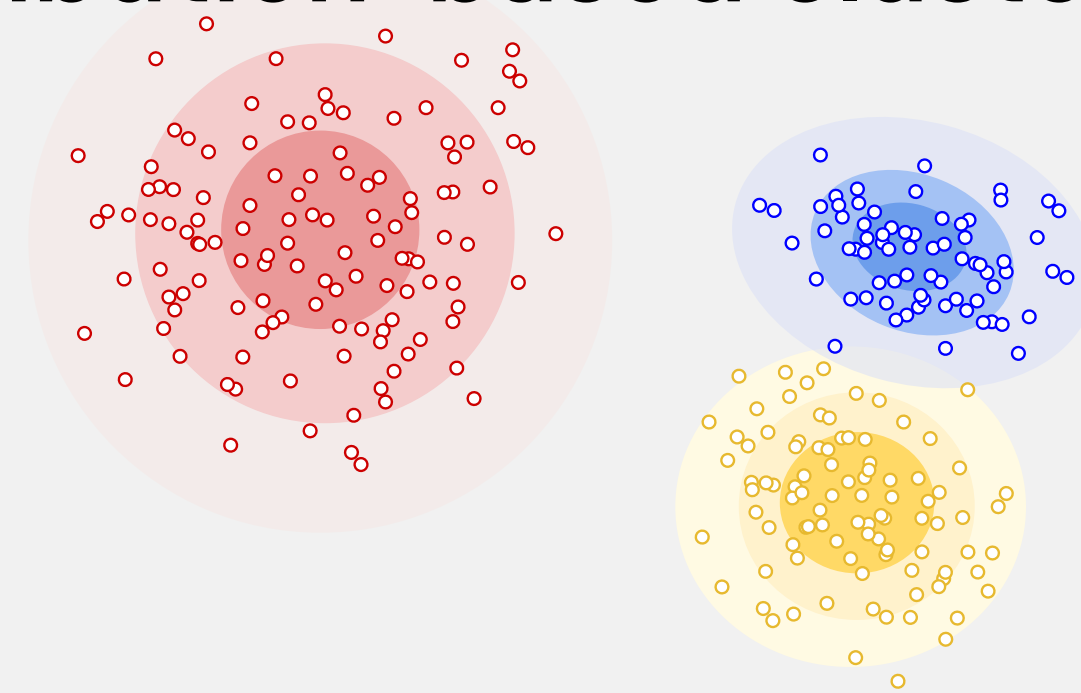
<https://developers.google.com/machine-learning/clustering/clustering-algorithms>

Q3 – Clustering Classification Algorithms

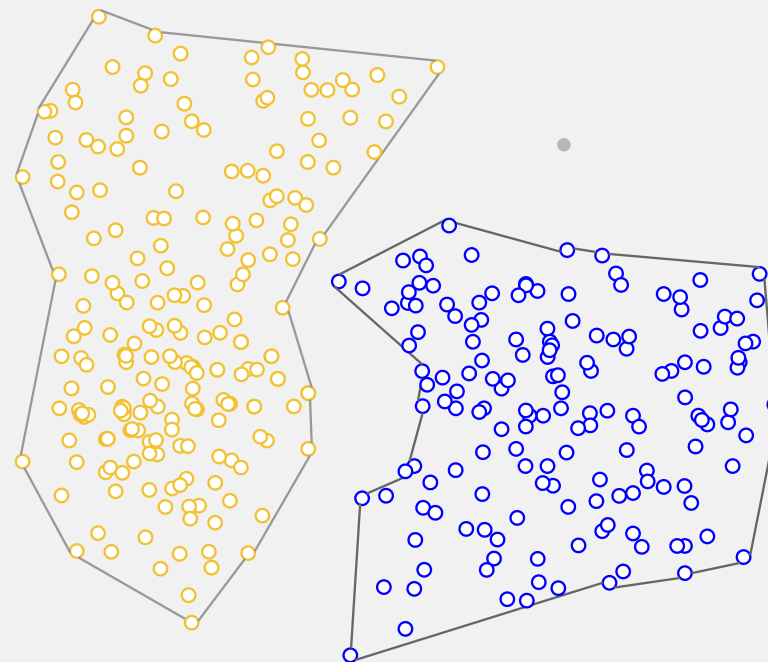
Centroid-based clustering



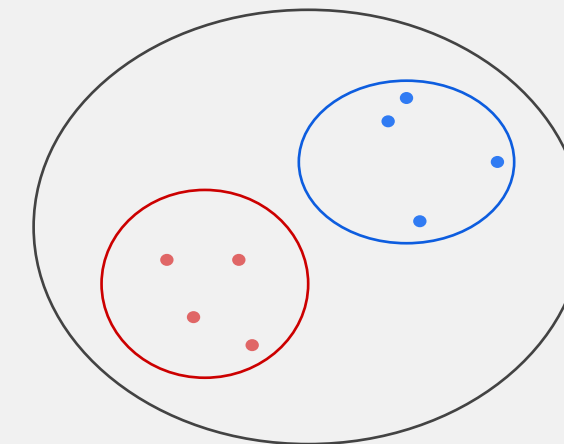
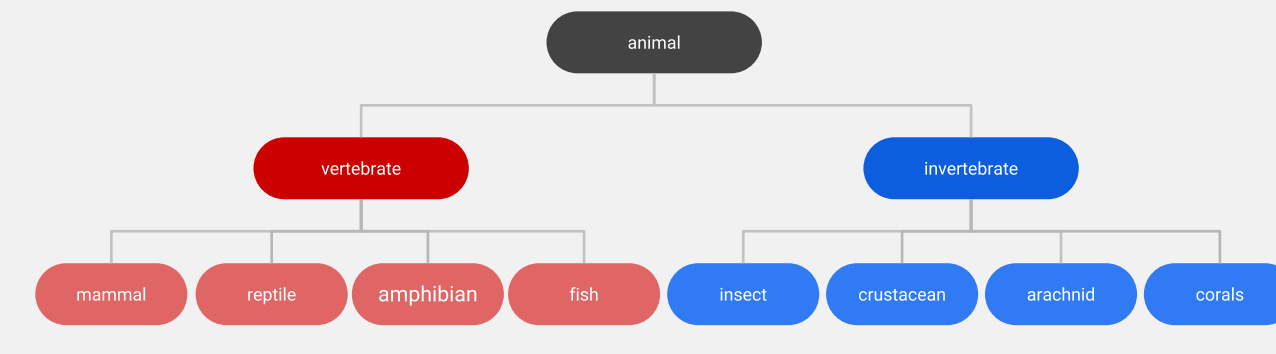
Distribution-based clustering



Density-based clustering



Hierarchical clustering



Source:

<https://developers.google.com/machine-learning/clustering/clustering-algorithms>

Q4 – Steps Involved in Building a Classification Model

Step	Explanation
1. Data Collection	Gather relevant data from various sources. This data should include the features (inputs) and the known class labels (outputs).
2. Data Preprocessing	Clean the data by handling missing values, removing noise, and normalizing or transforming variables for consistency.
3. Data Splitting	Divide the dataset into training and testing (and sometimes validation) sets. This allows you to train and then evaluate the model's performance.
4. Model Selection	Choose the appropriate classification algorithm (e.g., Decision Tree, Naive Bayes, SVM) based on data type and problem nature.
5. Model Training	Feed the training data to the algorithm so it can learn patterns and build the classification model.

Source:

Han, J., Pei, J., & Kamber, M. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann.

Q4 – Steps Involved in Building a Classification Model

Step	Explanation
6. Model Evaluation	Test the model on unseen data using metrics like accuracy, precision, recall, or confusion matrix to assess its performance.
7. Model Tuning	Fine-tune parameters (e.g., learning rate, depth) to improve accuracy and avoid overfitting or underfitting.
8. Deployment	Use the final model to make predictions on new data in a real-world setting.
9. Monitoring & Updating	Regularly monitor the model's performance over time and update it as needed to adapt to new data or changes in the environment.
10. Model Evaluation	Test the model on unseen data using metrics like accuracy, precision, recall, or confusion matrix to assess its performance.

Source:

Han, J., Pei, J., & Kamber, M. (2011). Data Mining: Concepts and Techniques (3rd ed.). Morgan Kaufmann.

Q5 – Common Evaluation Metrics Used in Classification

- Classification Accuracy
- Logarithmic Loss
- Confusion Matrix (Futher explained in Q9)
 - Accuracy
 - Precision or Positive Predictive Value (PPV)
 - Recall
 - F1-Score
- Receiver Operating Characteristic (ROC) Curve
- Area Under the Curve (AUC)

Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q5 – Common Evaluation Metrics Used in Classification – Classification Accuracy

The **ratio** of the **number of correct predictions (N)** to the **total number of predictions made (T)** for a dataset

$$Accuracy = \frac{N}{T}$$

Accuracy **works well for balanced classes** but is unreliable for imbalanced ones.

Why?

In a dataset with 100 samples, where 98 belong to class A and 2 to class B, a model that always predicts class A would achieve 98% accuracy, despite never predicting class B correctly.

Source:

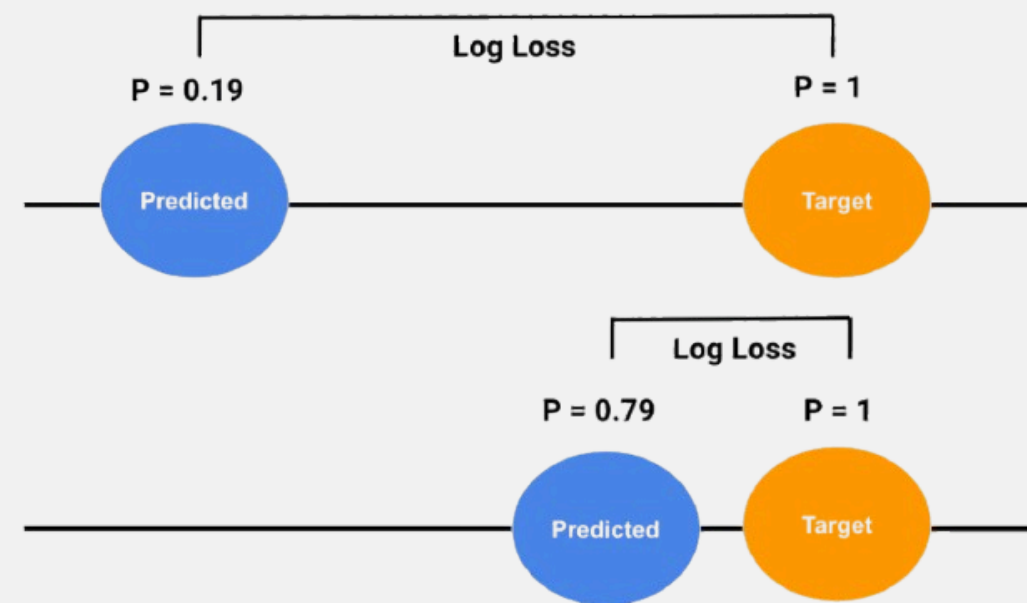
<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q5 – Common Evaluation Metrics Used in Classification – Logarithmic Loss / Log Loss

Measures the unpredictability of the extra noise that comes from using a predictor as opposed to the true labels

Use Case

The **output** of the classifier is a **numeric probability** instead of a class label



Poor prediction: Large difference between the actual and predicted probability → large log loss

Good Prediction: Predicted probability close to the actual probability → small log loss

Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q5 – Common Evaluation Metrics Used in Classification – Confusion Matrix

Variables

N – number of observations

M – number of possible class labels

y – a binary indicator (0 or 1) of whether sample i belongs to class j

p – the model's predicted probability of sample i belonging to class j

Binary Classification ($M = 2$)

$$\text{LogLoss} = - (y \log(p) + (1 - y) \log(1 - p))$$

Multiclass Classification ($M > 2$)

$$\text{LogLoss} = \frac{-1}{N} \sum_{i=1}^N \sum_{j=1}^M y_{ij} \log(p_{ij})$$

Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q5 – Common Evaluation Metrics Used in Classification – Confusion Matrix / Error Matrix

A table that shows the number of correct and incorrect predictions made by the model compared with the actual classifications in the test set

Evaluation Metrics Deatiled in
Q9

		Actual Value	
		Positives	Negatives
Predicted Value	Postives	True Positive (TP) Number of outcomes that are actually positive and are predicted positive.	False Positive (FP) Number of outcomes that are actually negative but predicted positive. These errors are also called Type 1 Errors .
	Negatives	False Negative (FN) Number of outcomes that are actually positive but predicted negative. These errors are also called Type 2 Errors .	True Negative (TN) Number of outcomes that are actually negative and are predicted negative.

Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

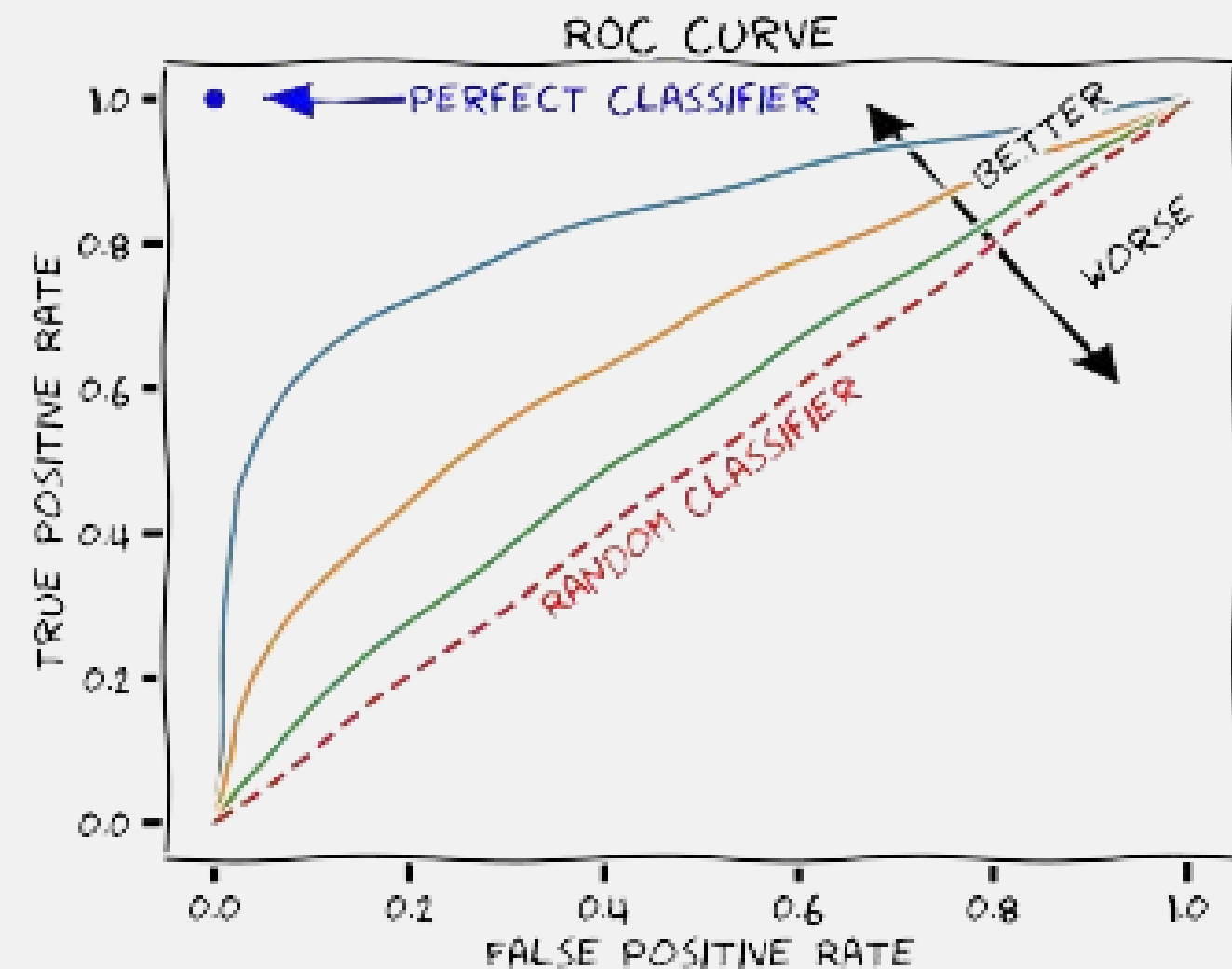
Q5 – Common Evaluation Metrics Used in Classification – Receiver Operating Characteristic (ROC) Curve

It plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various classification thresholds, showing the tradeoff between the two.

As the threshold decreases, the model classifies more observations as positive, which increases both TPR and FPR.

$$TPR = \frac{TruePositive}{FalseNegative + TruePositive}$$

$$FPR = \frac{FalsePositive}{TrueNegative + FalsePositive}$$



Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q5 – Common Evaluation Metrics Used in Classification – Area Under the ROC Curve (AUC)

AUC represents the probability that the classifier ranks a random positive example higher than a random negative example. It is a useful metric for comparing different models because it **summarises the entire ROC curve into a single value**.

- A value of 1 indicates perfect classification performance.
- A value of 0.5 indicates random guessing (no discriminatory ability).
- A value < 0.5 indicates worse than random guessing, which typically means that the model's predictions are inverted

The **higher the AUC, the better the model's ability** to distinguish between positive and negative classes across all thresholds.

Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

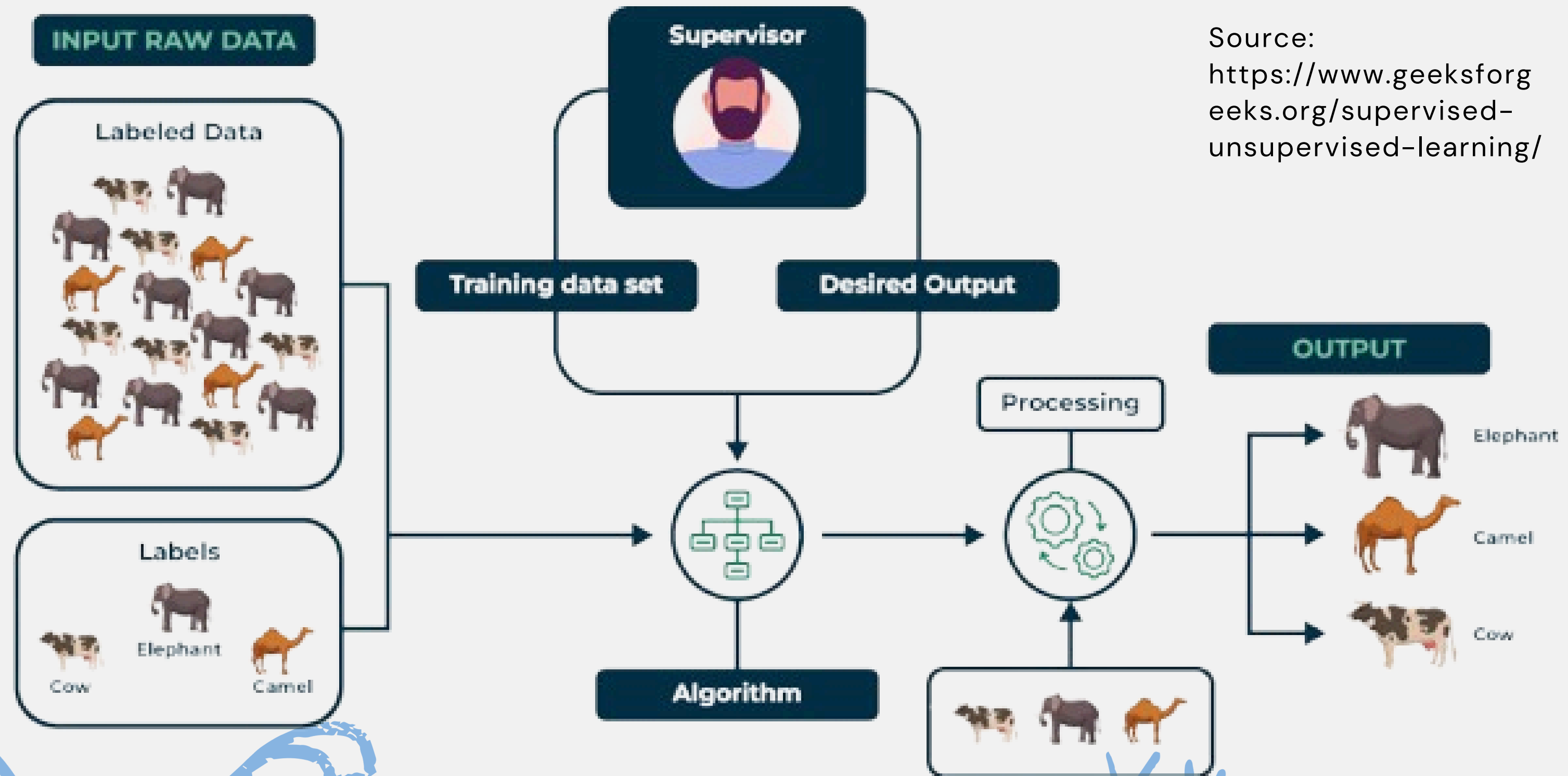
Q6 – Supervised Learning vs Unsupervised Learning

	Supervised Learning	Unsupervised Learning
Input Data	Algorithms are trained using labeled data	Algorithms are used against data that is not labeled
Accuracy	Highly accurate	Less accurate
Types	Regression, Classification	Clustering, Association
Application	Spam filtering, Image Classification, Medical diagnosis, Fraud Detection, NLP	Anomaly Detection, Scientific Discovery, Recommendation systems, Customer segmentation

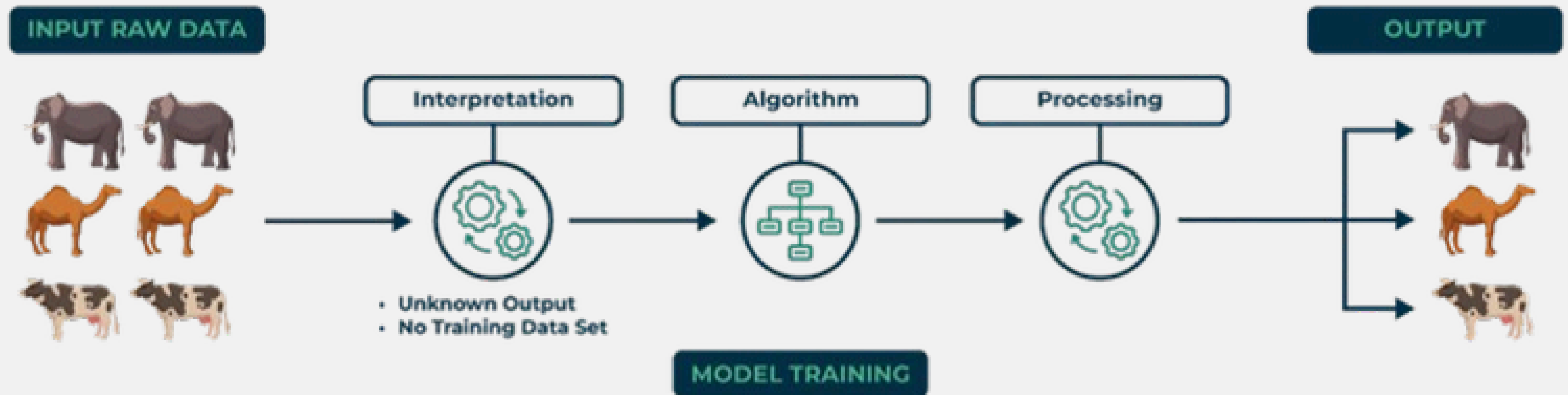
Source:

<https://www.geeksforgeeks.org/supervised-unsupervised-learning/>

Q6 – Supervised Learning



Q6 – Unsupervised Learning



Source:
<https://www.geeksforgeeks.org/supervised-unsupervised-learning/>

Q7 – Binary Classification vs Multi-class Classification

	Binary Classification	Multi-class Classification
No. of classes	2	More than 2
Algorithms	Logistic Regression k-Nearest Neighbors Decision Trees Support Vector Machine Naive Bayes	k-Nearest Neighbors Decision Trees Naive Bayes Random Forest Gradient Boosting
Examples	Email spam detection (spam or not) Churn prediction (churn or not)	Face classification Plant species classification

Source:

<https://www.analyticssteps.com/blogs/binary-and-multiclass-classification-machine-learning>

Q8 – Common Challenges in Classification Tasks

Challenges	Description	Solution
Data Imbalance	one class may be significantly more frequent than another, leading to biased models	Oversampling Undersampling Class weighting
Overfitting	models that learn too well from training data might not generalize to new data	Regularisation techniques Cross-validation
Feature Selection	process of identifying and selecting the most relevant features from a dataset to improve model performance and reduce overfitting	Recursive Feature Elimination (RFE) Feature importance ranking

Source:

<https://mljourney.com/understanding-classification-problems-in-machine-learning/>

Q9 – Confusion Matrix & Classification Model Evaluation

A table that shows the number of correct and incorrect predictions made by the model compared with the actual classifications in the test set

		Actual Value	
		Positives	Negatives
Predicted Value	Postives	True Positive (TP) Number of outcomes that are actually positive and are predicted positive.	False Positive (FP) Number of outcomes that are actually negative but predicted positive. These errors are also called Type 1 Errors .
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Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q9 – Confusion Matrix & Classification Model Evaluation

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision / Positive Predictive Value (PPV)

$$Precision = \frac{TP}{TP + FP}$$

Recall / Sensitive / True Positive Rate (TPR)

$$Recall = \frac{TP}{TP + FN}$$

F1-Score / F-measure

$$F1 = \frac{2}{\frac{1}{Precision} + \frac{1}{Recall}}$$
$$= \frac{2 \times precision \times recall}{precision + recall}$$

F-score with factor β

$$F_{\beta}Score = \frac{(1 + \beta^2) \times Precision \times Recall}{(\beta^2 \times Precision) + Recall}$$

$\beta < 1 \rightarrow$ Precision oreinted evaluation

$\beta > 1 \rightarrow$ Recall oreinted evaluation

Source:

<https://medium.com/analytics-vidhya/evaluation-metrics-for-classification-models-e2f0d8009d69>

Q10 – Overfitting & Underfitting

Source:

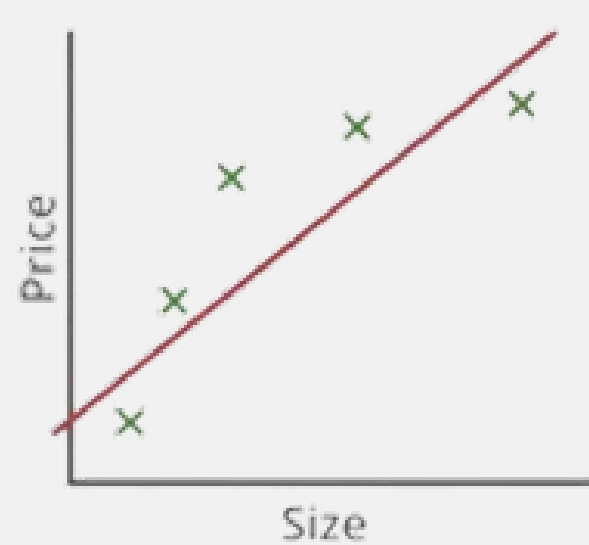
<https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>

	Overfitting	Underfitting
Situation	when a model learns too much from the training data, including details that don't matter (like noise or outliers)	when a model is too simple to capture what's going on in the data
Reasons	High variance and low bias The model is too complex	High bias and low variance The model is too simple
Performance	performing well on training data but poorly on test data	poor performance on both training and test sets
Bias-variance tradeoff	Increasing model complexity reduces bias but increases variance (risk of overfitting)	Simplifying the model reduces variance but increases bias (risk of underfitting)

Q10 – Overfitting & Underfitting

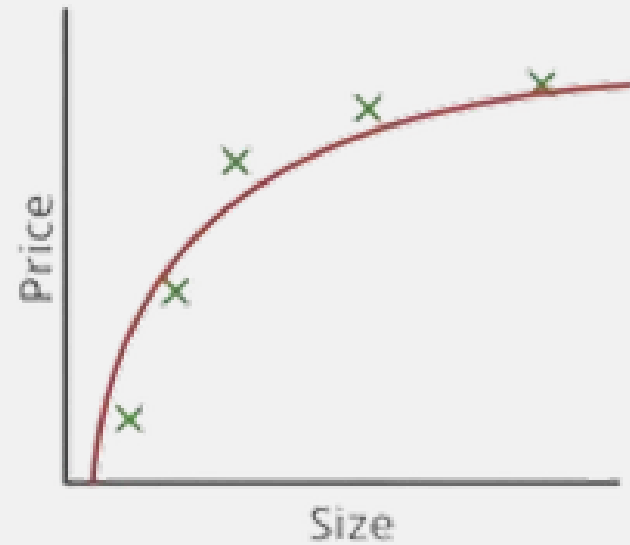
Source:

<https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>



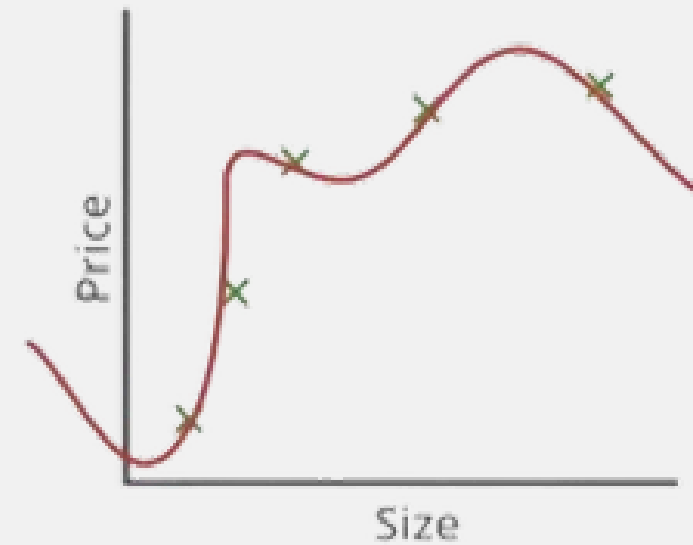
$$\theta_0 + \theta_1 x$$

High Bias
(Underfitting)



$$\theta_0 + \theta_1 x + \theta_2 x^2$$

Low Bias, Low Variance
(Goodfitting)



$$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$$

High Variance
(Overfitting)

Underfitting

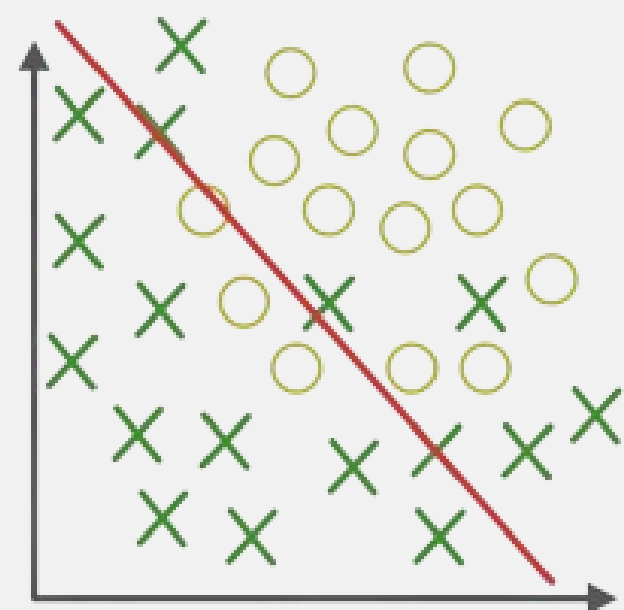
Straight line trying to fit a curved dataset but cannot capture the data's patterns

Overfitting

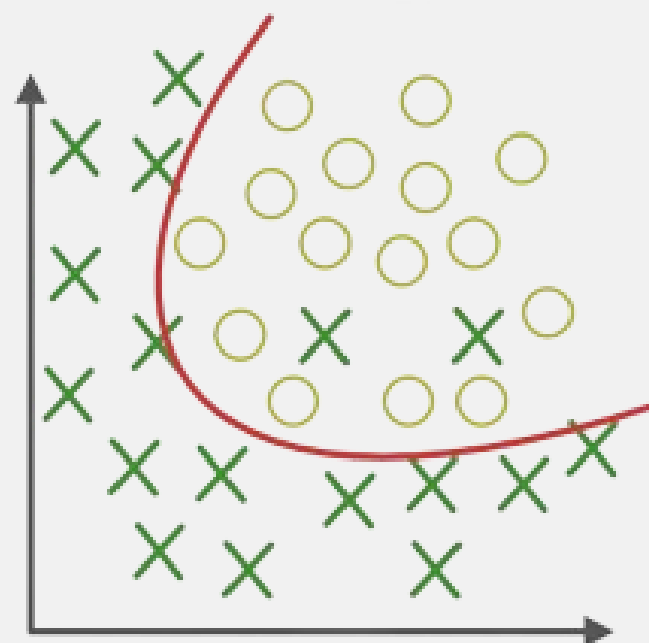
A squiggly curve passing through all training points, failing to generalise

Appropriate Fitting

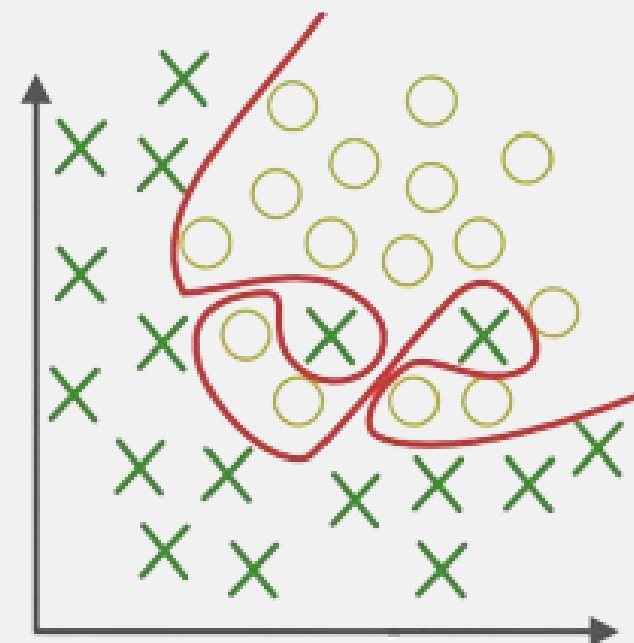
Curve that follows the data trend without overcomplicating to capture the true patterns in the data



Under-fitting
(too simple to explain the variance)



Appropriate-fitting



Over-fitting
(forcefitting--too good to be true)

Q10 – Overfitting & Underfitting

	Overfitting	Underfitting
Solution	Remove noise from the data	Remove noise from the data
	Reduce model complexity	Increase model complexity
	Early stopping during the training phase (as soon as loss begins to increase stop training)	Increase the number of epochs or increase the duration of training to get better results
	Ridge Regularisation Lasso Regularisation	Feature Engineering

Source:

<https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>

Q11 – Describe the role of cross-validation in evaluating the performance of a classification algorithm.

Role of cross-validation is to estimate the skill of a ML model by **partitioning the data into multiple subsets**. It helps *mitigate issues like overfitting and underfitting* by evaluating the model's performance on different subsets of the dataset. Useful for small datasets where a single train-test split may not provide a reliable performance measure.



k-fold cross-validation

Data is split into k subsets, and the model is trained on $k-1$ of these subsets, testing it on the remaining subset. This process is repeated for each fold, and the overall performance is averaged to provide a robust estimate of model accuracy.

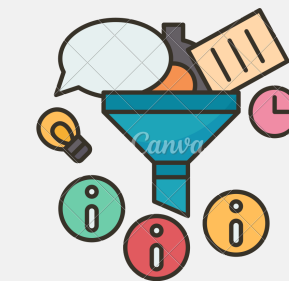


Q12 – How can feature selection improve classification model performance? Mention at least two methods.

Feature selection improves the classification model's performance by reducing the dimensionality of dataset, removing irrelevant and redundant features that will interrupt model's learning process. Two common methods:

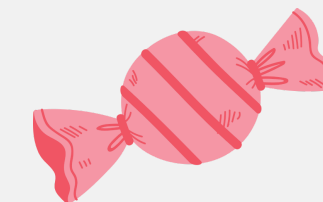
- **Fileter Method**

This involves selecting features based on their statistical properties with the target variable.

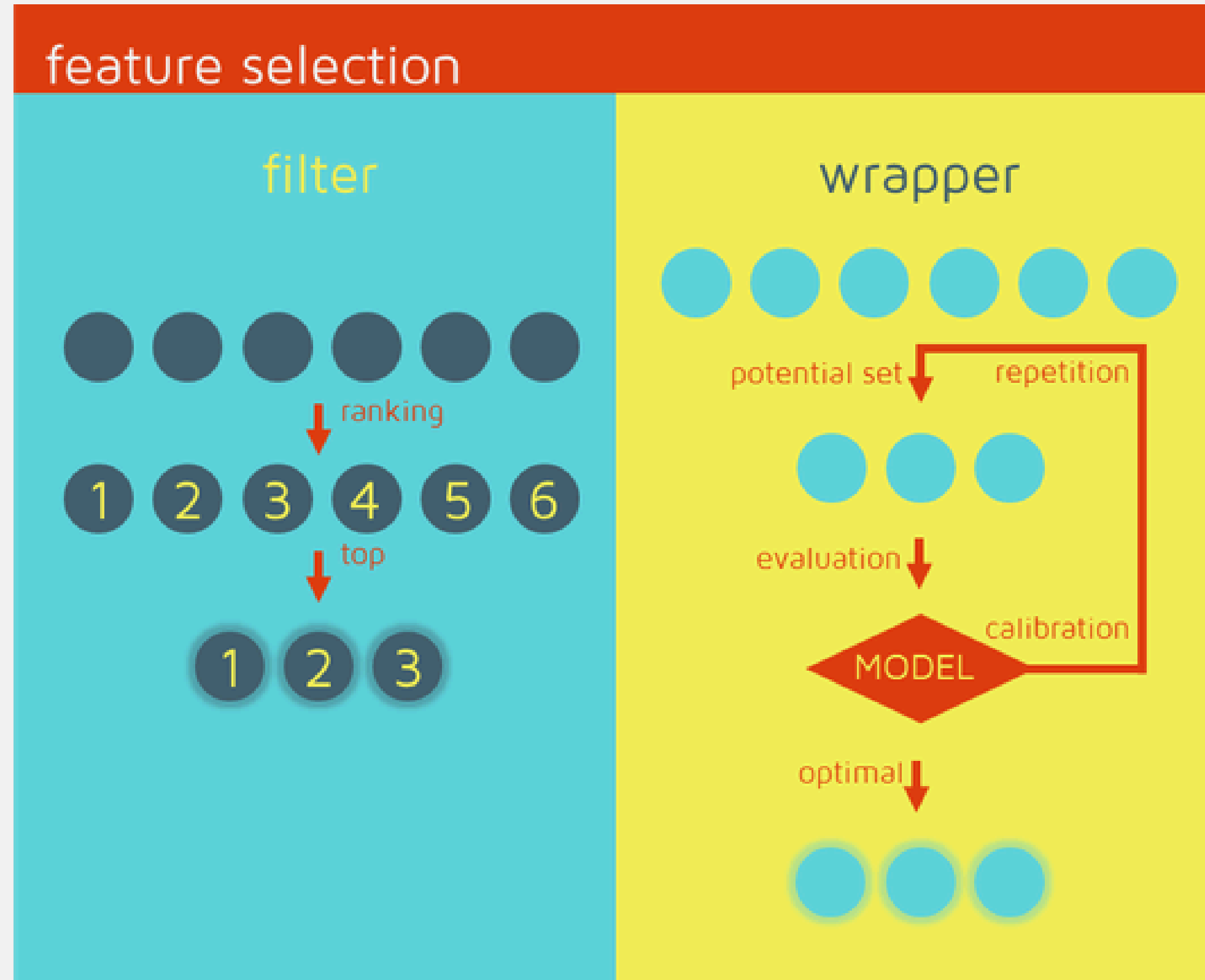


- **Wrapper Method**

Algorithms like recursive feature elimination (RFE) iteratively remove features and evaluate model accuracy, selecting the best-performing subset of features



Q12 – How can feature selection improve classification model performance? Mention at least two methods.

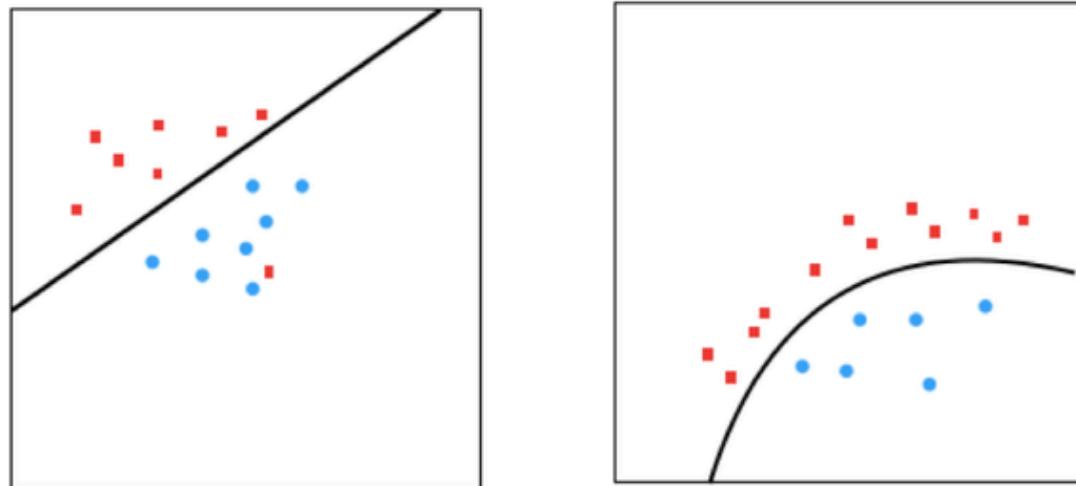


Q13 – Compare decision trees and logistic regression in terms of model interpretability, accuracy, and use cases.

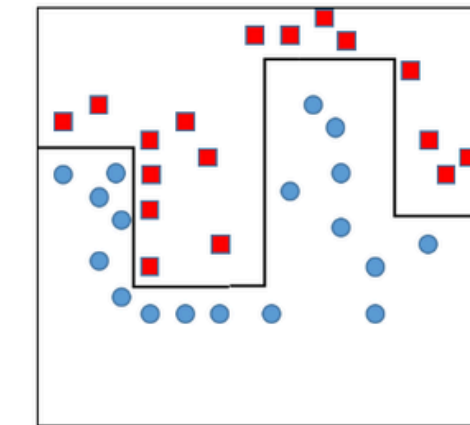
Aspect	Decision Trees	Logistic Regression
Interpretability	Highly interpretable; graphical structure allows easy understanding of decision-making process	Interpretable but less intuitive, requires understanding of the relationship between features and log-odds
Accuracy	Can overfit the data, especially with deep trees; performance may suffer with noisy data	Less prone to overfitting, but may underperform if the relationship is non-linear or complex
Use Cases	Ideal for problems with non-linear relationships or complex feature interactions (e.g., fraud detection, medical diagnosis)	Suitable for problems with linear relationships between features and target (e.g., predicting customer churn, disease risk)

Q13 – Compare decision trees and logistic regression in terms of model interpretability, accuracy, and use cases.

Logistic Regression assumes that the data is linearly (or curvy linearly) separable in space.



Decision Trees are non-linear classifiers; they do not require data to be linearly separable.



Non-linearly separable data

Linearly non - separable?	Decision Tree
Categorical Data type	Decision Tree
Continuous Data type	Logistic Regression
Skewed	Decision Tree, or give high weight to minority class in Logistic Regression
Outlier	Decision Tree, or remove outlier for Logistic Regression
Lots of Missing values	Decision Tree

Q14– Explain what ensemble methods (e.g., Bagging, Boosting, Random Forest) are and how they improve classification performance.

- **Bagging**

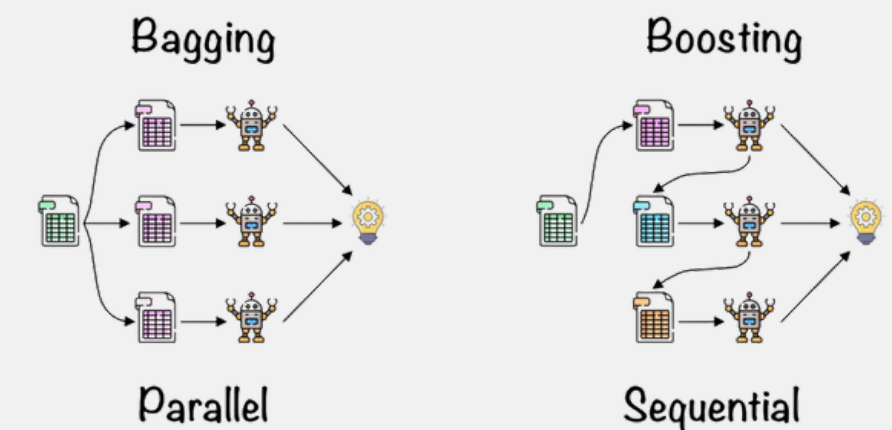
This technique generates multiple subsets of the training data using random sampling with replacement and trains a classifier on each subset. The final prediction is made by averaging the predictions of all classifiers (for regression) or by majority voting (for classification).

Exp : *Random Forest*

- **Boosting**

Enhance the sequential construction of models, where each new model will correct the errors of the previous model. It focuses on the training instances that were wrongly classified by the early models and assigns them more weights.

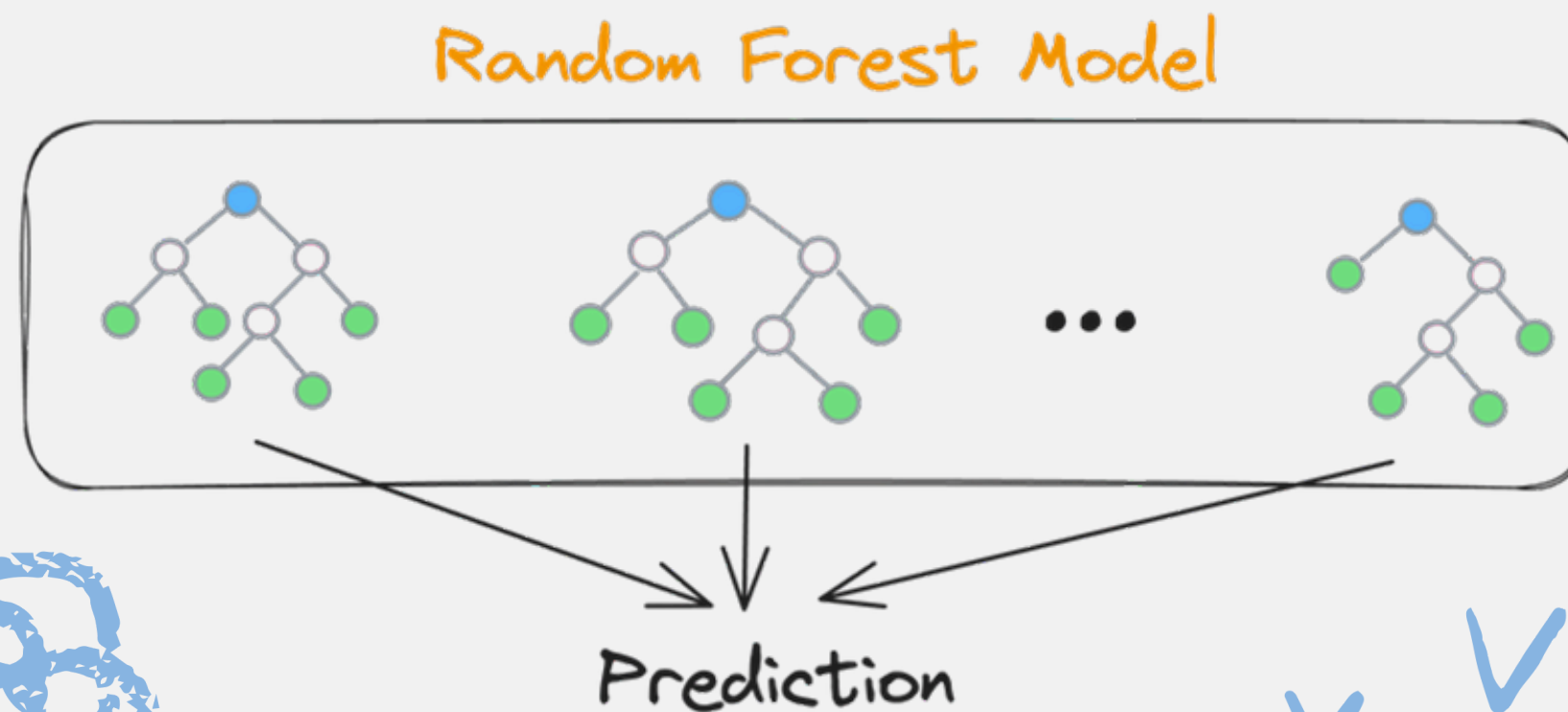
Exp : *Gradient Boosting*



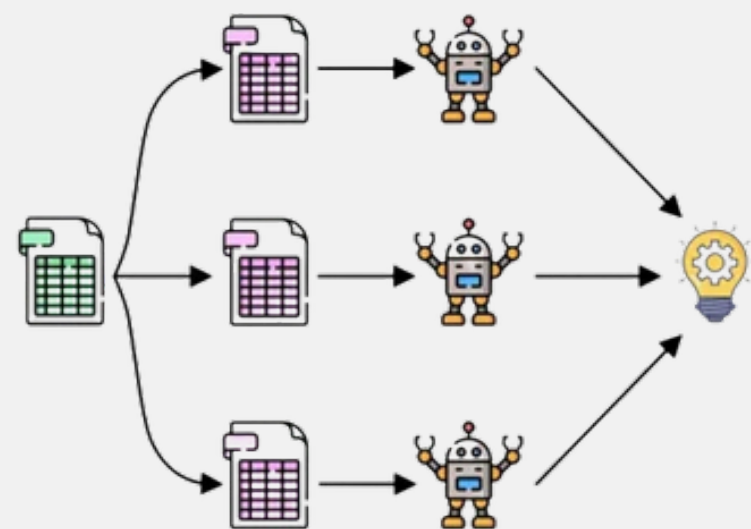
Q14- Explain what ensemble methods (e.g., Bagging, Boosting, Random Forest) are and how they improve classification performance.

- **Random Forest**

An integrated method combining bagging and decision trees. It builds multiple decision trees using different random data subsets and features, and makes the final classification decision by summarizing the results of all the trees. It is robust to overfitting and provides better accuracy than a single decision tree

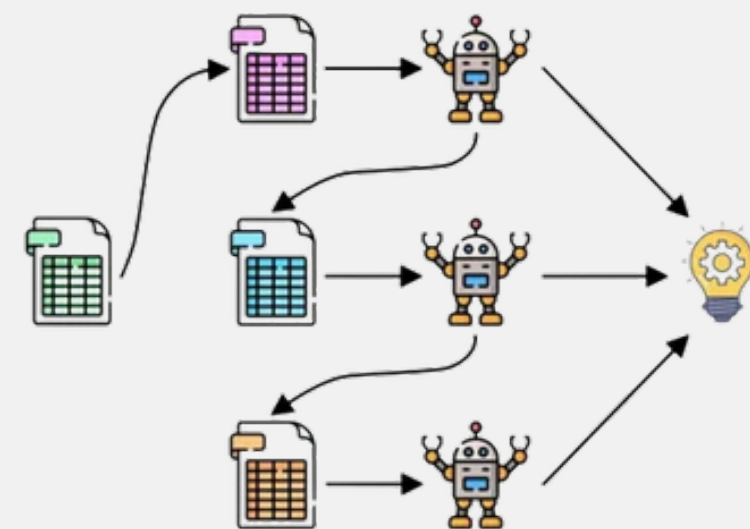


Bagging



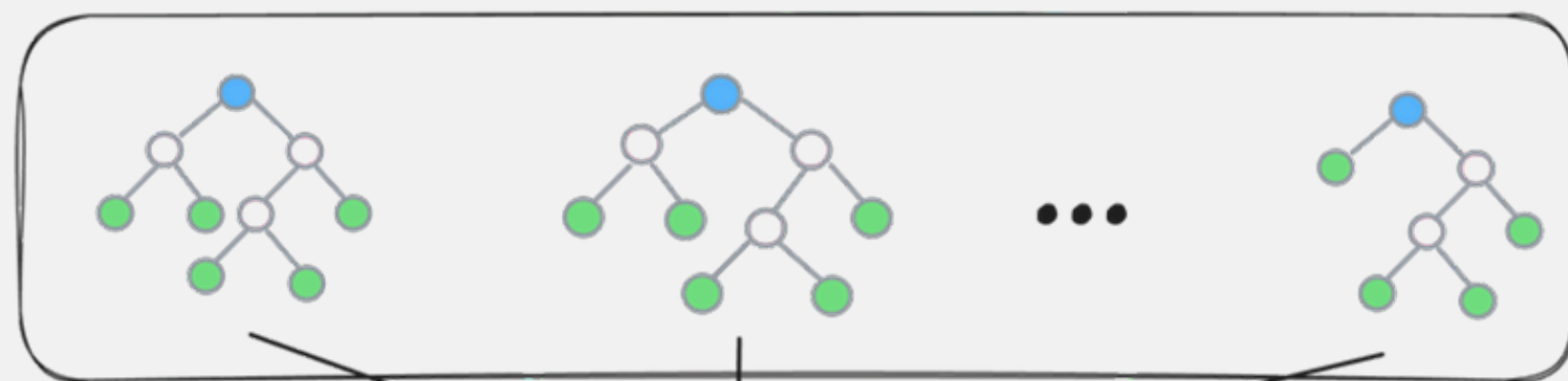
Parallel

Boosting



Sequential

Random Forest Model



Prediction

The background is a light gray color, decorated with various hand-drawn blue doodles. These include several overlapping circles and loops at the top, a series of concentric arcs at the bottom left, a wavy line at the bottom center, and several small 'v' shapes at the bottom right. On the far right edge, there are some vertical blue strokes. The central text is in a large, bold, black sans-serif font with a white drop shadow.

**Thank you
very much!**